

How to Read Enrichment Dot Plots

diabetes-cerna-rnaseq · a learner's guide (GO + KEGG, Day 8)

Part 1 – What a dot plot actually is

After differential expression you have a long list of **changed** genes. Enrichment asks one question: **do these genes share a common job?** The **dot plot is the standard way to display that answer.**

The mental model: each row is a *biological function*, and the dot tells you **three** things at **once** about how strongly your gene list hits that function.

A dot plot packs **four** **channels of information** into one picture:

Channel	What it shows	How to read it
Y-axis (rows)	the enriched terms/pathways	usually sorted with the strongest at top
X-axis (GeneRatio)	what fraction of your genes are in this term	further right = bigger share of your list
Dot size (Count)	how many of your genes are in this term	bigger dot = more genes
Dot colour (p.adjust)	statistical confidence	red = more significant, blue = less

Read them in that order: **term -> position -> size -> colour.**

Part 2 – The **three** numbers, explained properly

GeneRatio (the x-axis)

Written as a fraction like **60/455**. It means: *of the 455 genes from my list that could be tested, 60 of them belong to this term*. Plotted as the decimal ($60/455 = 0.13$).

- **Further right = this term captures a bigger slice of your gene list.**
- It does **not** mean “more important” on its own – always read it alongside colour (significance).

Count (the dot size)

The raw number of your genes in that term – the numerator of GeneRatio (60).

Why have both Count and GeneRatio? **Count is absolute, GeneRatio is relative.** A term with 60 genes out of 455 (**13%**) is a stronger **hit** than a term with 60 genes out of **5,000** (**1.2%**). Size shows the raw hit; position shows the proportion.

p.adjust (the colour) – the one people **misread**

Step 1: what a p-value is. It answers “**what’s the chance I’d see this much overlap purely by luck?**” **Small p = unlikely to be luck** = probably real.

Step 2: why it must be *adjusted*. GO tests **thousands** of terms at once. If you test **5,000** terms at **$p < 0.05$** , roughly **250** will look “**significant**” by pure chance. That’s the multiple-testing problem. **p.adjust** (here using the **Benjamini-Hochberg method**) corrects for this, so the number you see already accounts for all those tests. **Always judge by p.adjust, never raw p-value.**

Step 3: reading the scale. The legend shows scientific notation:

Written	Means	Reading
1e-03	0.001	1 in 1,000
1e-06	0.000001	1 in a million
1e-12	0.000000000001	1 in a trillion
1e-16	0.0000000000000001	1 in 10 quadrillion

Smaller number = more significant. So **1e-16** is vastly stronger evidence than **1e-03**. In clusterProfiler’s default colouring, **red = smaller p.adjust (more significant)** and **blue = larger p.adjust (less significant)**.

THE TRAP: colour scales are **per-plot**, not absolute

This catches almost every beginner. Each dot plot rescales its **own colours** to its own range of p-values. A red dot in one figure is **not** comparable to a red dot in another.

In your own figures: - **Up plot:** colour scale runs 1e-16 (red) to 1e-10 (blue). - **Down plot:** colour scale runs 1e-06 (red) to 1e-03 (blue).

So the reddest dot on the down plot (1e-06) is **far weaker evidence** than the **bluest dot on the up plot (1e-10)** – a **thousand times weaker**. **Never compare colours across panels without checking both legends first.**

Part 3 – What “good” looks like, and what to worry about

Signs of a healthy, informative dot plot:

- **Terms tell a coherent story.** They point at one or two related biological themes, not a random scatter of unrelated functions.
- **Strong significance.** Top terms at $1e-08$ or smaller are solid; $1e-15$ and beyond is very strong.
- **Reasonable Counts.** Terms containing 10-60 of your genes are meaningful. Terms with 2-3 genes are fragile – easily driven by chance.
- **Distinct terms.** Each row says something different.
- **Readable labels.** No overlaps, no truncation.

Red flags:

- **Massive redundancy** – 12 rows all saying “telomere maintenance” in slightly different words. This is GO’s parent/child hierarchy leaking through, and it crowds out real findings. **Fix:** `simplify()` (see Part 4).
- **Tiny counts everywhere** (Counts of 2-4) – likely noise.
- **Weak significance** (p.adjust near 0.05 throughout) – a borderline result.
- **Incoherent terms** with no biological through-line – possibly a bad gene list.
- **A term you can’t explain** – investigate before reporting it.

Part 4 – The redundancy problem (and why simplify matters)

GO is a **hierarchy**: “T cell activation” sits inside “lymphocyte activation”, inside “leukocyte activation”, inside “immune system process”. When your genes hit one, they hit all of them – so enrichment reports the whole family.

The consequence: your first down-plot spent **12 of 20 rows** on near-identical telomere/Cajal-body terms. Real, distinct findings were pushed off the figure.

The fix:

```
ego_s <- simplify(ego, cutoff = 0.7, by = "p.adjust", select_fun = min)
```

This measures **semantic similarity** between terms and, for any cluster more than **70%** similar, keeps only the most significant representative.

This is not cosmetic. In your data, simplifying the **down-regulated** set revealed **tricarboxylic acid cycle**, **Golgi vesicle transport**, **Golgi organization**, and **regulation of translation** – terms that were previously **buried**, and which changed the biological interpretation entirely.

Rule: **always simplify GO output before drawing conclusions from it.**

Part 5 – Reading YOUR three plots

GO – up in T2D

- **Scale:** p.adjust 1e-16 to 1e-10 – **extremely strong**.
- **Counts:** up to **60** genes per term – substantial.
- **GeneRatio:** up to 0.13, i.e. **13%** of your tested up-genes fall in the **top** term.

Themes: regulation of **leukocyte activation** (top, 1e-16), T cell activation, leukocyte cell-cell adhesion, leukocyte mediated immunity, **chemotaxis** and **leukocyte migration**, humoral immune response, **peptide antigen assembly with MHC class II**, and **extracellular matrix organization**.

Interpretation: immune cells are being actively **recruited** (chemotaxis, migration), **activated** (T cell activation), and **presenting antigen** (MHC class II) – alongside **fibrotic tissue remodelling** (ECM organization). This is not passive inflammation; it is an organised immune response plus scarring.

GO – down in T2D

- **Scale:** p.adjust 1e-06 to 1e-03 – real, but **orders of magnitude weaker than the up set**. Worth stating honestly.
- **Counts:** 10-40 genes.
- **GeneRatio:** 0.01-0.05 – smaller slices than the up plot.

Themes: ribosome biogenesis (top), regulation of translation, **Golgi vesicle transport**, rRNA metabolic process, protein folding, protein stabilization, **tricarboxylic acid cycle**, Golgi organization, telomere organization.

Interpretation – the beta-cell machinery story. Beta cells **sense glucose** through **mitochondrial metabolism** (glucose -> glycolysis -> **TCA cycle** -> ATP -> K-ATP channel closes -> calcium influx -> **insulin release**), and they manufacture insulin through **translation** -> **ER folding** -> **Golgi trafficking** -> **secretory granules**. Your down-regulated program hits **exactly those steps**. So this is **impaired glucose-stimulated insulin secretion**, described at the level of **the machinery that performs it** rather than the GO label “insulin secretion”.

Important nuance: the GO term “insulin secretion” itself is **not** enriched. Individual beta-cell genes (**SLC2A2/GLUT2, CLTRN**) are significantly **down**, but too **few** to drive pathway-level enrichment. Say it that way – precisely.

KEGG – DE mRNAs

The disease-name trap. Your KEGG list includes *Staphylococcus aureus infection*, *Leishmaniasis*, *Asthma*, *Autoimmune thyroid disease*, *Type I diabetes mellitus*, *Allograft rejection*, *Graft-versus-host disease*, *Systemic lupus erythematosus*.

These are **not diagnoses**. KEGG’s disease pathways are **built** from **heavily overlapping gene sets** – MHC/antigen presentation, cytokines, T-cell receptor signalling, complement. Any strong generic immune signature lights up **every** immune-related disease pathway simultaneously.

Correct reading: “a broad adaptive-immune and antigen-presentation program is activated.” **Incorrect reading:** “these islets show a Type 1 diabetes pathway.” The mechanistically informative

entries are the non-disease ones: *complement and coagulation cascades, hematopoietic cell lineage, cell adhesion molecules, Th1 and Th2 cell differentiation, integrin signalling.*

Part 6 – Your reusable checklist

Run through these **five** steps on any dot plot:

1. **Read both legends first** – note the p.adjust range and the Count range. Never compare colours across figures without doing this.
2. **Scan the terms for a theme** – is there a coherent biological story, or scattered noise?
3. **Check the top term’s numbers** – how significant, how many genes, what fraction of your list?
4. **Check for redundancy** – if several rows restate one idea, simplify and re-plot before concluding anything.
5. **Interrogate surprising terms** – especially disease names in KEGG. Ask *why* that term appeared before you report it.

Phrases to use (and avoid)

- **Say:** “Immune activation terms were strongly enriched (p.adjust $\sim 1e-16$).”
- **Say:** “Down-regulated genes were enriched for translation, Golgi transport and TCA cycle, consistent with reduced secretory capacity.”
- **Say:** “Individual beta-cell genes were down-regulated but did not reach pathway-level enrichment.”
- **Avoid:** “The Type I diabetes pathway was enriched” (shared gene sets, not a diagnosis).
- **Avoid:** comparing dot colours between two different figures.
- **Avoid:** reporting redundant parent/child terms as separate findings.